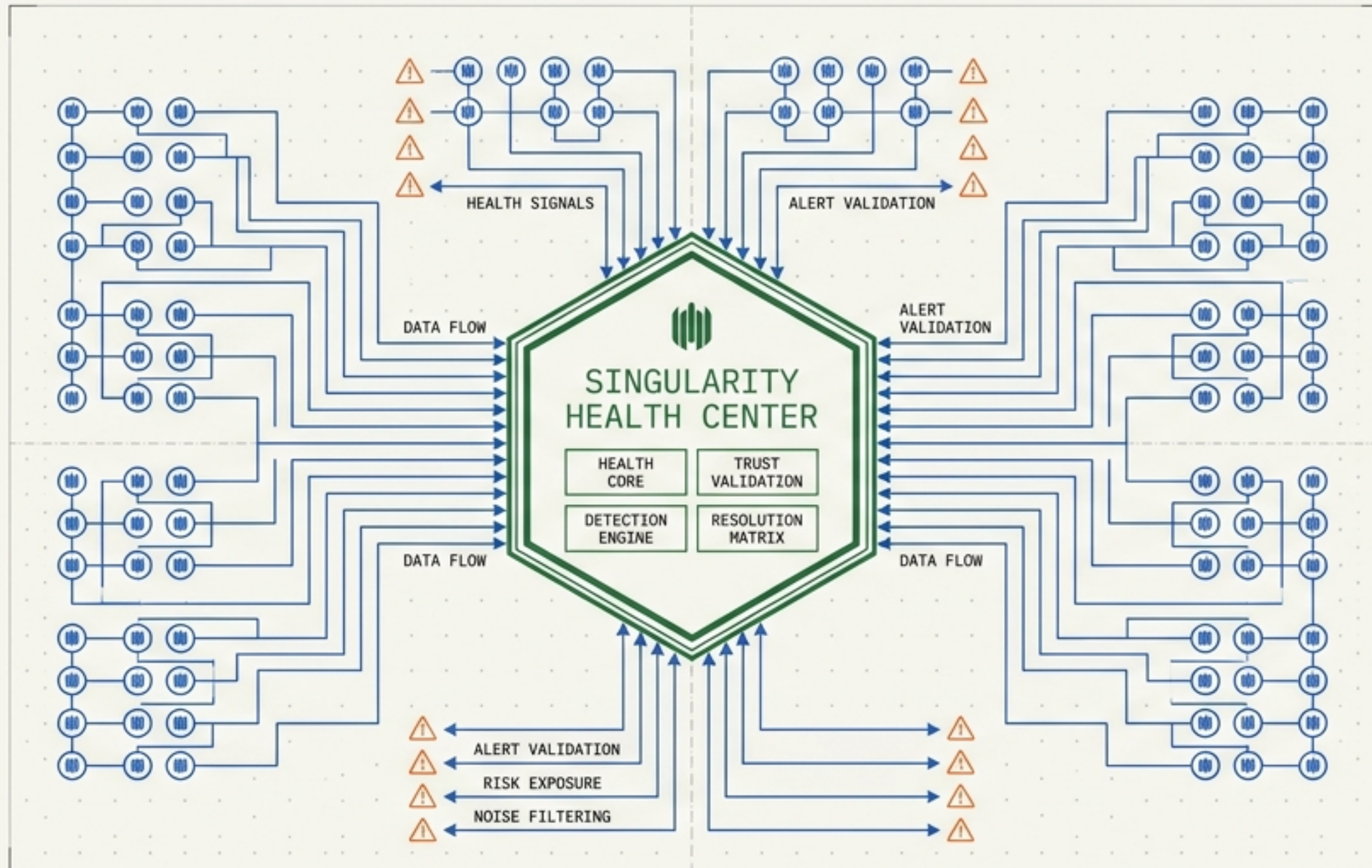


Blueprinting the Singularity Health Center

Delivering a hyperscale agent platform engineered to monitor, detect, and resolve operational health issues across millions of SentinelOne endpoints.



ENGINEERING OBJECTIVES

- HYPERSCALE MONITORING
- PRECISION DETECTION
- AUTONOMOUS RESOLUTION

OPERATIONAL METRICS

ENDPOINTS: 10+ MILLION
LATENCY: <5ms (P95)
DETECTION RATE: >99.99%

Bharat Verma

Senior Engineering Manager,
Agent Platform

Technical Review & Operating Playbook

Trust Requires Mathematical Promises

We are building an authoritative operational health platform. Missing data is a signal, but it is not proof of endpoint failure. We must mathematically differentiate between endpoint offline, network partition, and pipeline delay.

SCALE



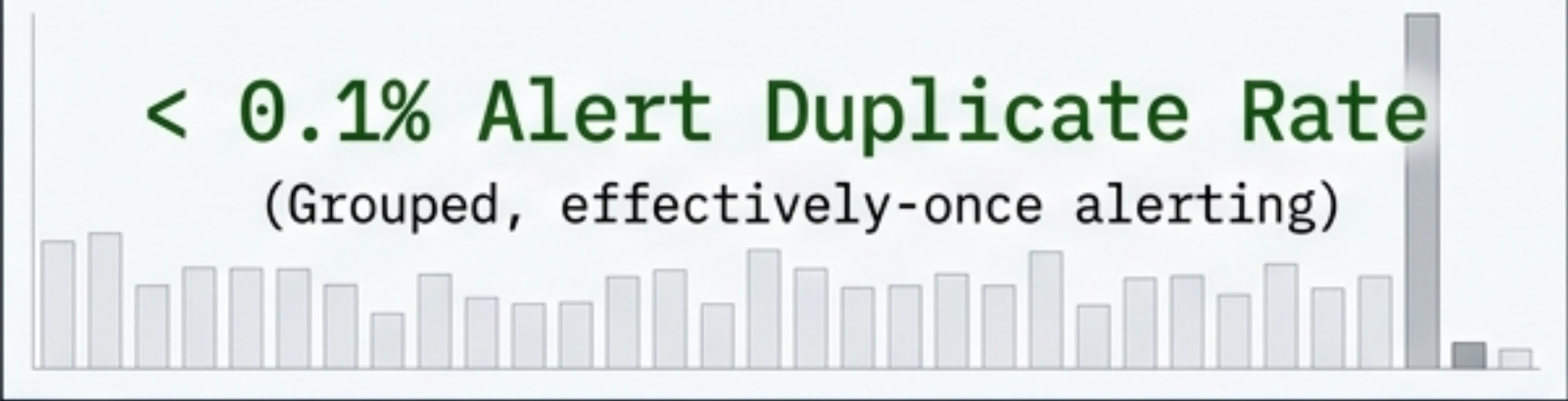
RESILIENCE



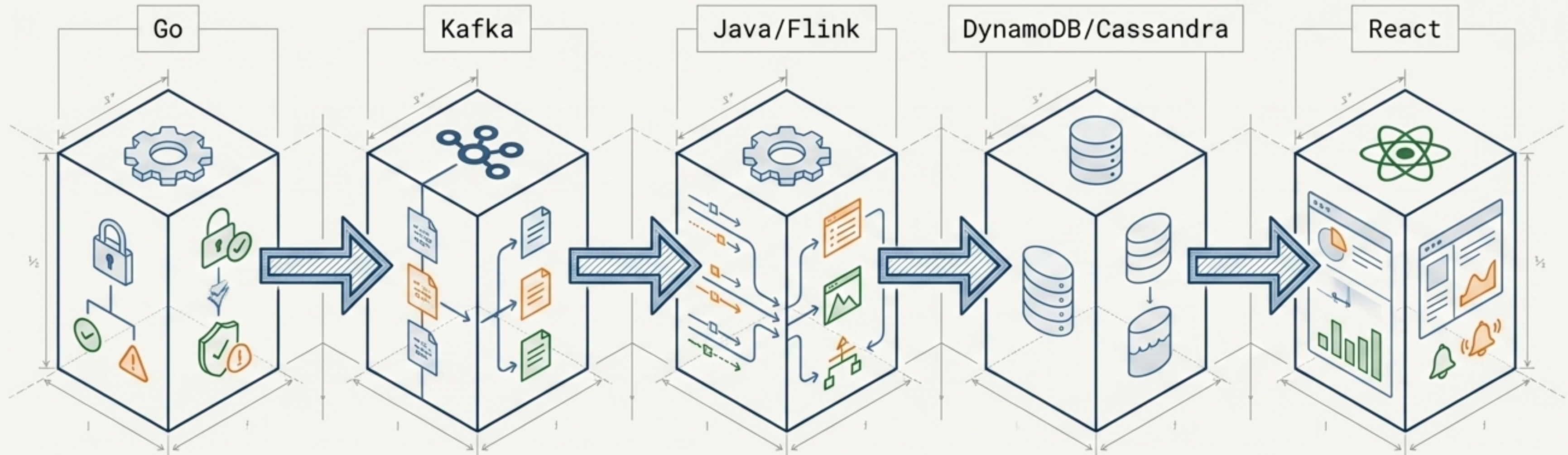
SPEED



PRECISION



End-to-End Streaming Backbone



1. Edge & Ingestion

Regional Gateways terminate auth and drop invalid schemas. Fast ACKs protect endpoints.

2. Streaming Backbone

Partitioned by `tenant_id` + `hash(agent_id)` to isolate noisy tenants while preserving order.

3. Processors & Rules

Normalization, deduplication, and event-time windowing. Applies versioned rules.

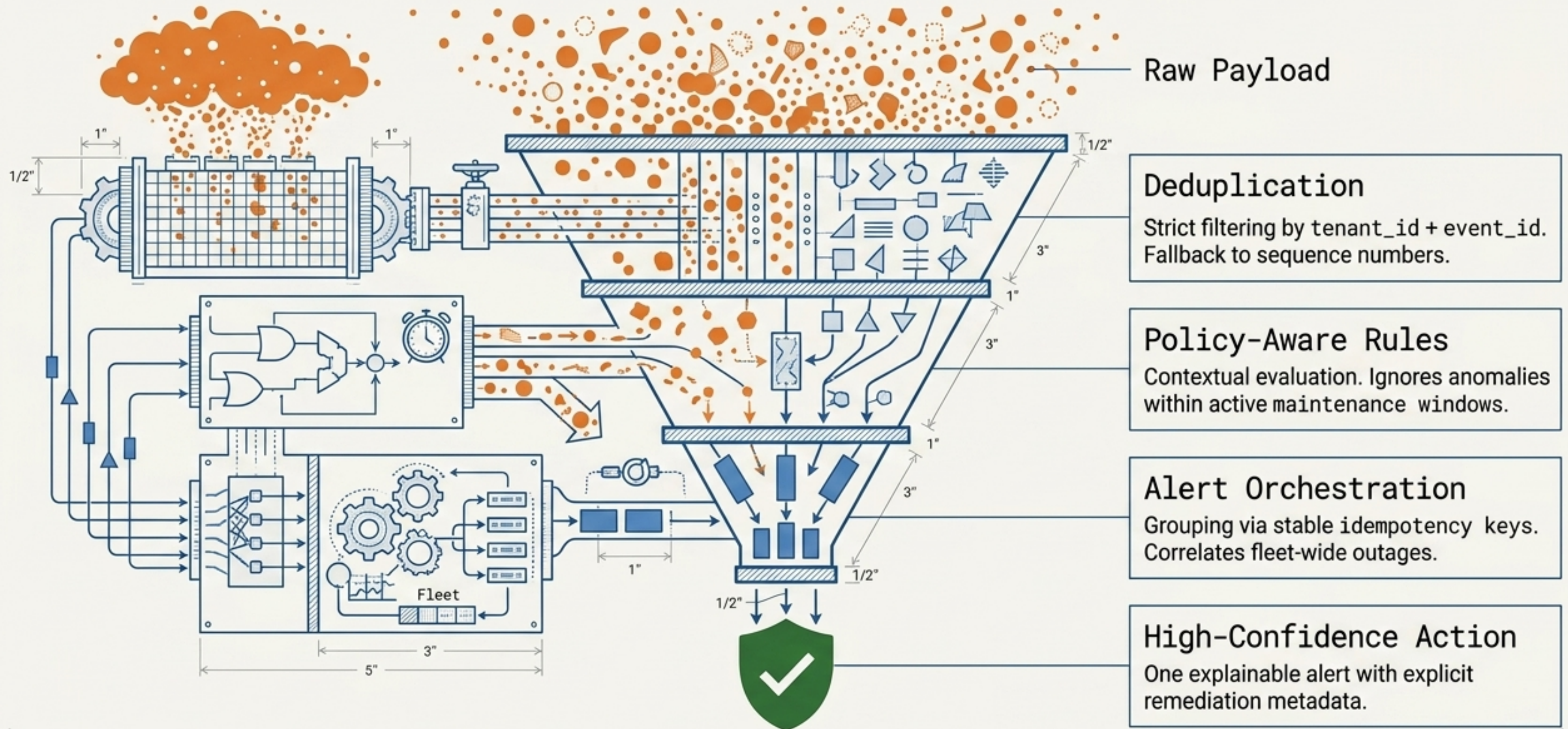
4. State & Storage

Hot state key-value store for authoritative latest health. Raw events to Data Lake.

5. Alerting & UI

Orchestration layer suppresses noise. React UI consumes pre-computed read models.

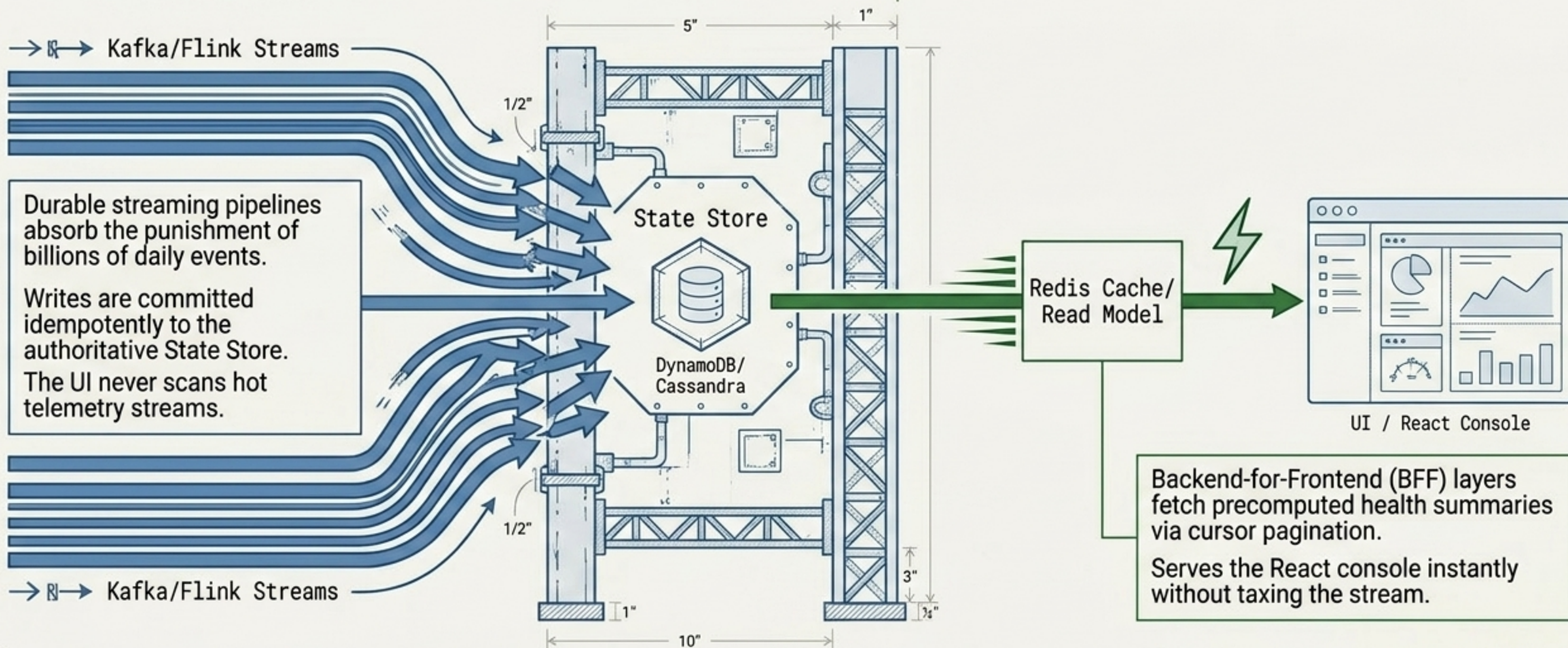
Protecting Customer Trust Over Alerting Volume



Isolating High-Volume Writes From Low-Latency Reads

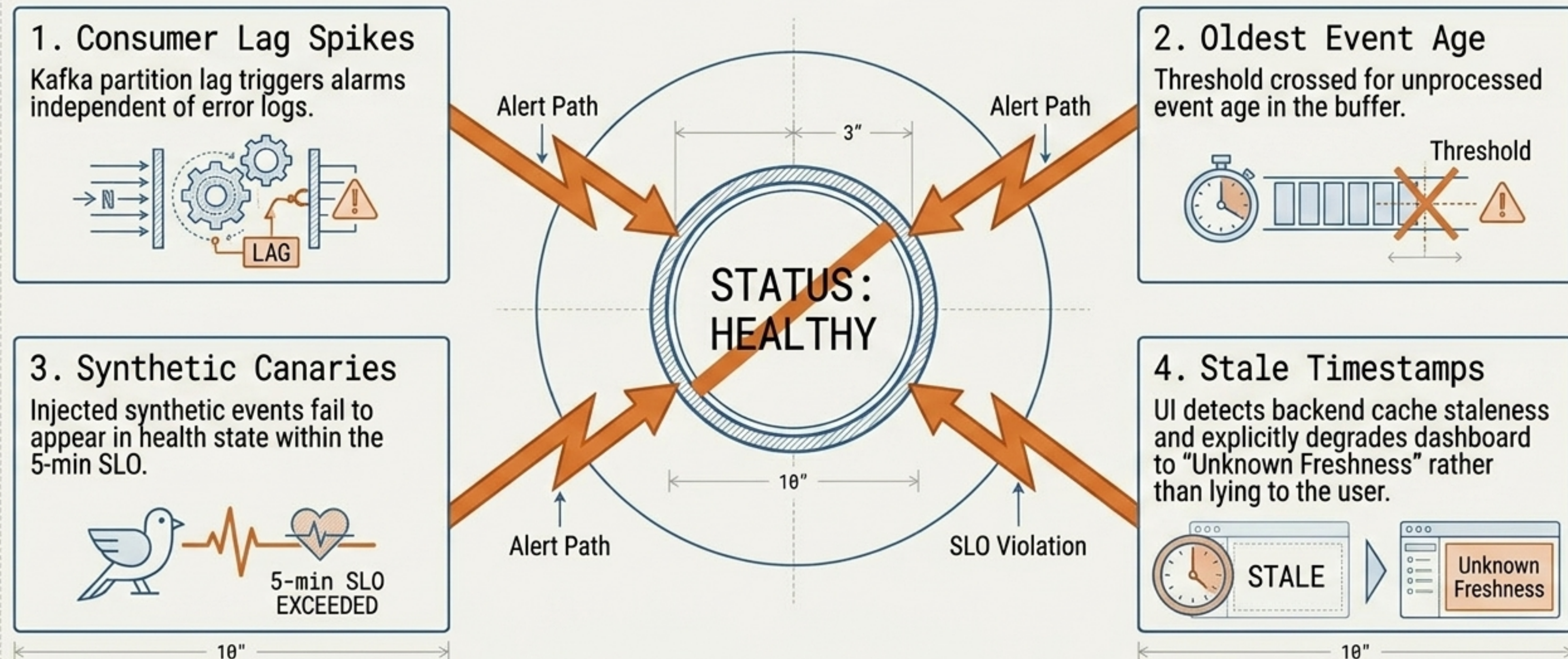
The Write Path: Massive Throughput

The Read Path: Sub-200ms Access



The Sev-1 Response: Surviving a Silent Stop

Scenario: Telemetry events **silently** fail to process. The UI looks completely healthy, creating a dangerous false sense of security for customers.



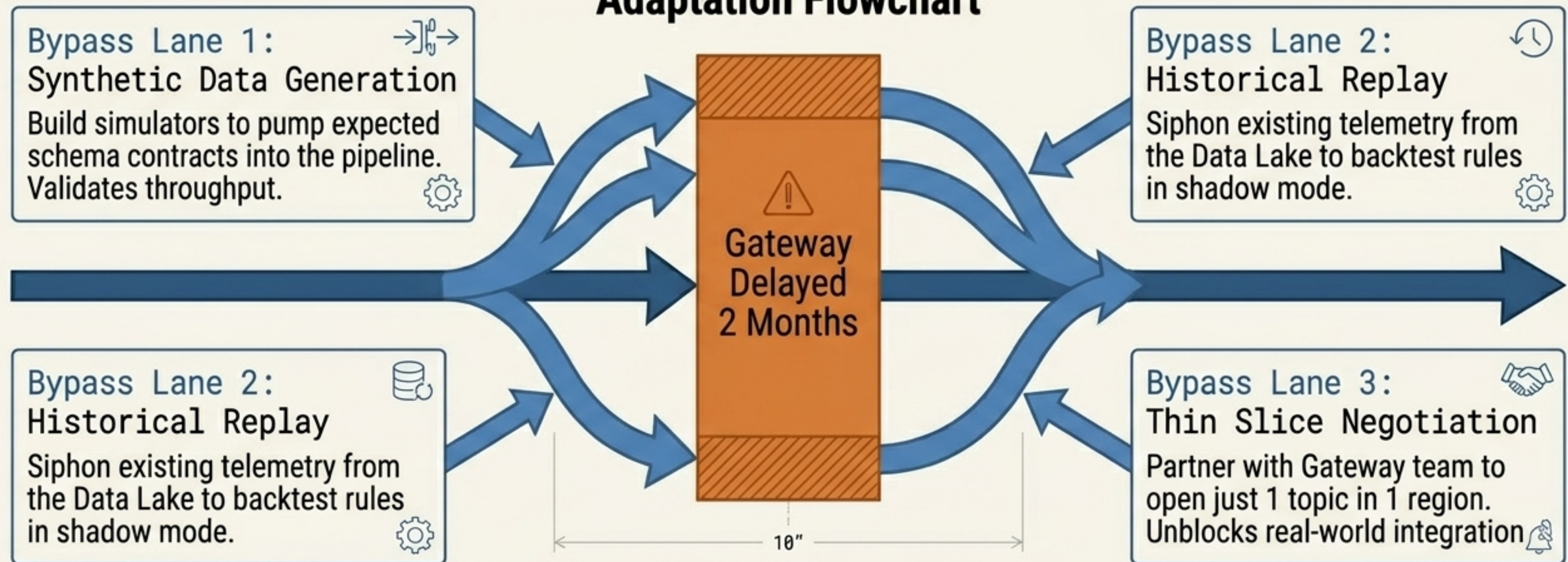
Pragmatic Phasing: Proving Value Before Proving Scale

Dimension	Minimum Viable Product (MVP)	General Availability (GA)
Primary Goal 	Prove core pipeline & establish observability 	Safely replace legacy health status globally 
Alert Routing 	Shadow mode execution, limited preview tenants 	Default enablement with broad notification fanout 
Rule Scope 	Core 4 rules (Tamper, Disabled, Disk, Heartbeat) 	Fleet-wide groupings, aggregations, and policy drift 
Exit Criteria 	5-min latency achieved, zero UI raw scans 	Sev-1 Game Days passed, legacy systems retired 

Executing Through Dependency Risk

The Threat: The core Ingestion Gateway team is delayed by 2 months. They cannot route new production health telemetry to our stream.

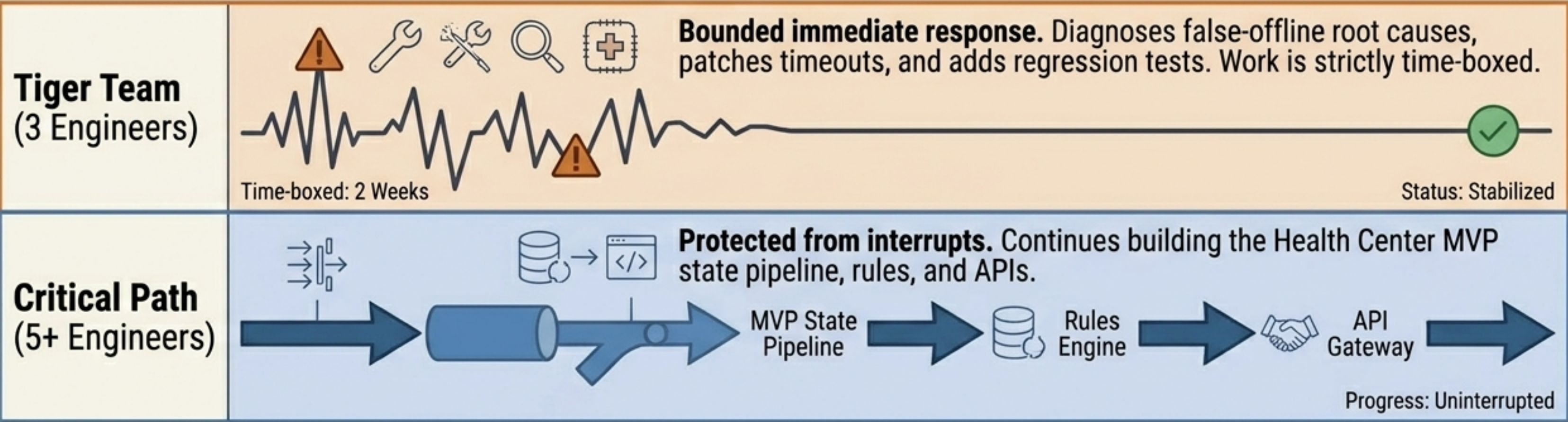
Adaptation Flowchart



Quarantining the Operational Drain

The Crisis: A 40% spike in customer escalations due to legacy heartbeat microservice showing false 'Offline' statuses.

Capacity Allocation Swimlanes

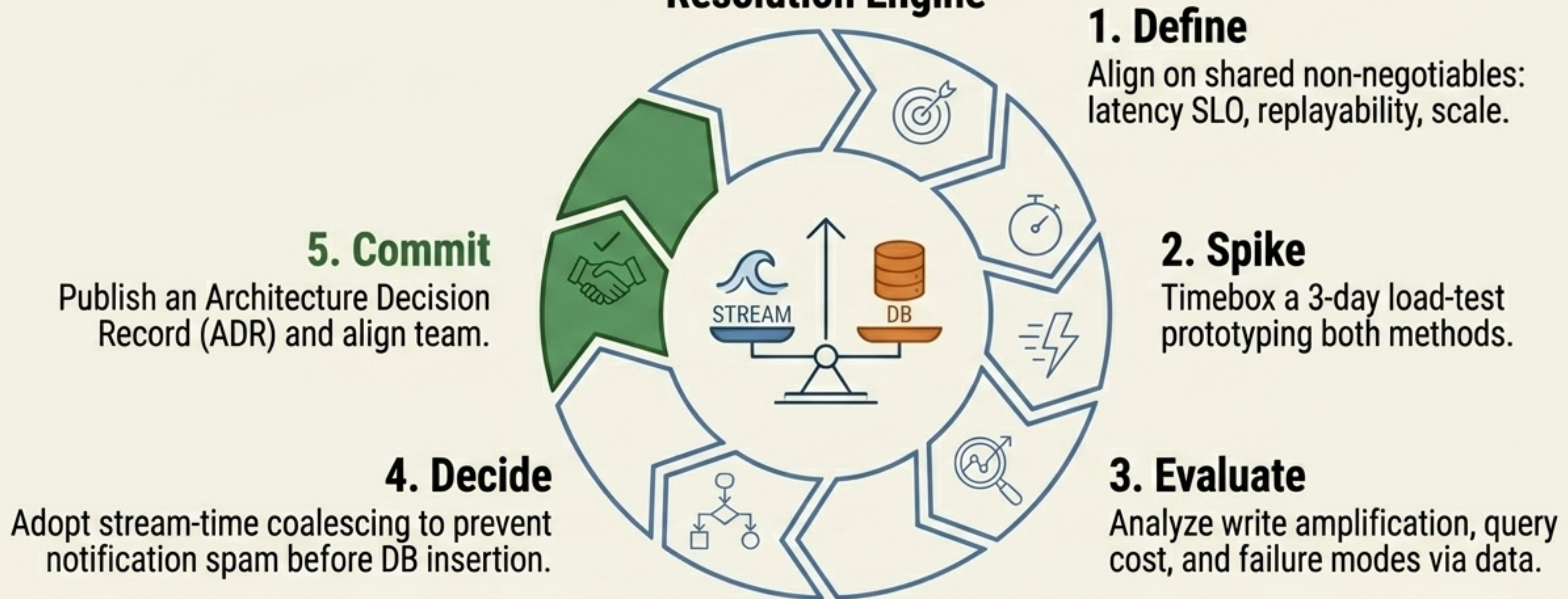


Strategic Outcome: We do not sacrifice current customer trust to ship the future, but we refuse to let legacy fires consume the entire team.

Resolving the Alert Coalescing Conflict

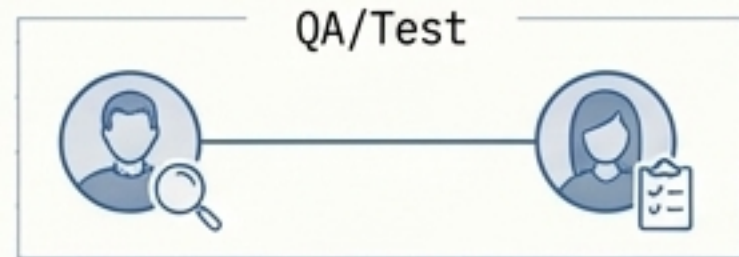
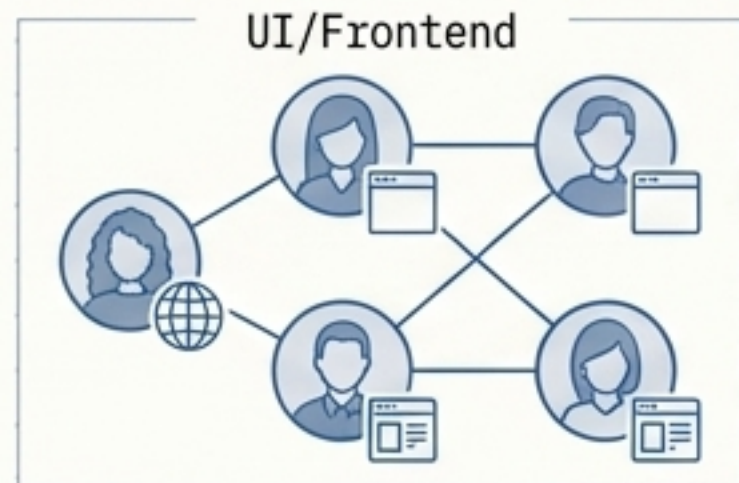
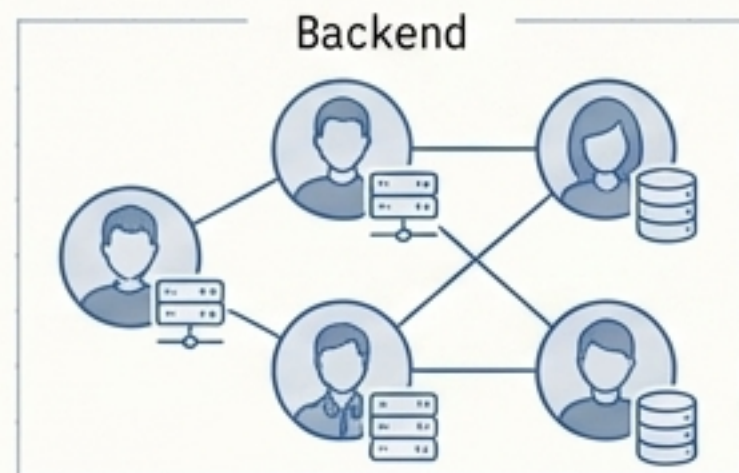
The Conflict: Engineer A argues for Stream Processing (optimizes for 5-min latency, high complexity). Engineer B argues for DB writes + Cron jobs (optimizes for simplicity, risks scale/delays).

Resolution Engine



Shielding Morale Through Ambiguity

The Challenge: Team morale suffers when engineers feel buffeted by unpredictable legacy issues and undefined work, leading to burnout and attrition.



Visible Reality

Maintaining a **transparent roadmap**. By explicitly tracking dependency blocks and legacy stabilization as **planned work**, we eliminate ambient anxiety.

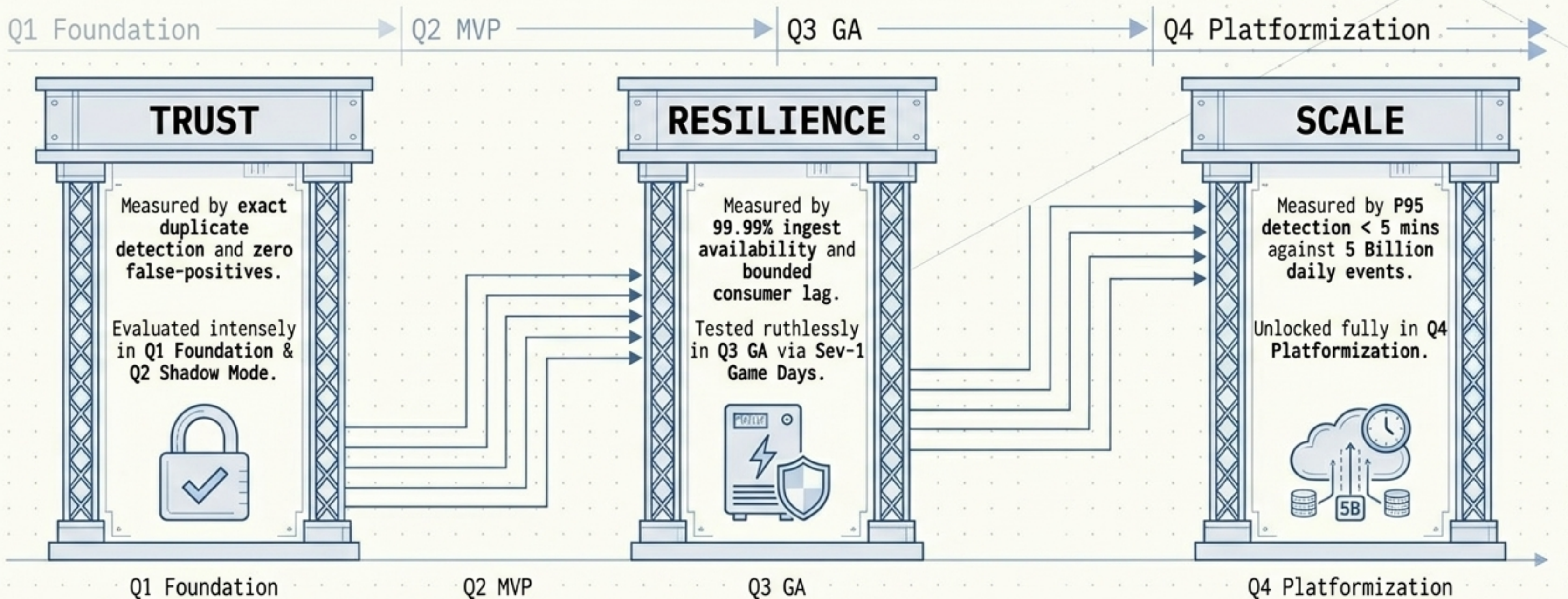
Bounded Interruptions

Rotating **incident commanders**. Engineers know exactly when they are the shield (handling the legacy spike) and when they are protected.

Meaningful Agency

Turning operational fires into growth opportunities. Senior engineers drive **ADRs**; QA leads **replay automation**. Progress is celebrated.

Singularity Health Center: The Path Forward



Operational health products live or die by trust. We isolate ingestion from detection, prove value in shadow mode, and relentlessly protect our team's focus to deliver a hyperscale platform that never lies to the customer.